UCSD Spring 2026 DSC148: Data Mining

DSC 148: Intro to Data Mining – Data Mining Challenge

Overview

Overview

In this competition, you will build a machine learning model to predict the nightly price of short-term rental listings in New York City.

Each listing includes structured information about the property, host, location, booking rules, amenities, reviews, and several free-text description fields. Your task is to use these listing attributes to predict the `price` column for listings in the test set.

This is a supervised regression problem. The training data includes the observed nightly price for each listing, while the test data includes the same listing features without the target value.

The dataset is intentionally realistic: it contains missing values, categorical variables, date fields, currency-like strings, sparse columns, text fields, and a skewed target distribution. A successful solution will require thoughtful preprocessing, feature engineering, model selection, and evaluation.

Start
10 days ago

Close
12 days to go

Description

Goal

Predict the nightly rental price for each listing in `test.csv`.

Your submitted file should contain one prediction for each test listing, using the required format:

```
Id,Predicted
19307997,145.177291430616
20176193,145.177291430616
```

Dataset Files

File	Description
`train.csv`	Training data with listing features and the target column `price`.
`test.csv`	Test data with the same listing features, excluding `price`.
`mean_value_baseline.csv`	Baseline submission that predicts the mean training price for every test listing.
`simple_linear_regression_baseline.csv`	Baseline submission generated from a simple linear regression model.
`simple_baseline.ipynb`	Starter notebook showing how to load the data and create baseline predictions.

Data Summary

The training set contains 33,538 listings, and the test set contains 17,337 listings.

The target variable is:

```
price
```

The training price distribution is skewed, with the following summary:

Statistic	Value
Mean	145.18
Median	109.00
Minimum	0.00
25th percentile	69.00
75th percentile	175.00
Maximum	1,999.00

Because prices are skewed and contain large outliers, you may want to consider transformations such as `log(price + 1)` during model training. If you transform the target, remember to convert predictions back to dollar values before submitting.

Feature Types

The dataset contains a mixture of feature types.

Listing Text Fields

These columns contain natural-language descriptions of the listing, space, rules, neighborhood, and host interaction.

Examples include:

```
name
summary
space
description
neighborhood_overview
notes
transit
access
interaction
house_rules
```

These fields can be used directly with text models or converted into engineered features such as text length, keyword indicators, TF-IDF features, or embeddings.

Host Information

These columns describe the host account and host behavior.

Examples include:

```
host_id
host_name
host_since
host_location
host_about
host_response_time
host_response_rate
host_is_superhost
host_neighbourhood
host_listings_count
host_verifications
host_has_profile_pic
host_identity_verified
```

Some host fields contain missing values or string formatting that should be cleaned before modeling.

Location Fields

These columns describe where the listing is located.

Examples include:

```
neighbourhood_cleansed
neighbourhood_group_cleansed
city
state
zipcode
market
country_code
country
```

The listings are primarily from New York City boroughs, including Manhattan, Brooklyn, Queens, the Bronx, and Staten Island. Location is likely to be an important predictor of price.

Property and Room Attributes

These columns describe the rental unit itself.

Examples include:

```
property_type
room_type
accommodates
bathrooms
bedrooms
beds
bed_type
amenities
square_feet
```

Common values include entire homes or apartments, private rooms, and shared rooms. Numeric capacity-related features such as `accommodates`, `bedrooms`, `bathrooms`, and `beds` are likely to be useful predictors.

Booking and Policy Fields

These columns describe booking rules and guest requirements.

Examples include:

```
guests_included
extra_people
minimum_nights
maximum_nights
instant_bookable
is_business_travel_ready
cancellation_policy
require_guest_profile_picture
require_guest_phone_verification
```

Some of these fields may be boolean-like strings or currency-like strings and may require conversion before they can be used in most machine learning models.

Review Fields

These columns describe the quantity and quality of guest reviews.

Examples include:

```
number_of_reviews
first_review
last_review
review_scores_rating
review_scores_accuracy
review_scores_cleanliness
review_scores_checkin
review_scores_communication
review_scores_location
review_scores_value
reviews_per_month
```

Review fields may be missing for listings with few or no reviews. Missingness itself may be informative.

Modeling Suggestions

A basic workflow might include:

1. Load `train.csv` and `test.csv`.
2. Separate the target variable `price` from the training features.
3. Clean numeric fields stored as strings, such as currency or percentage fields.
4. Impute missing values for numeric and categorical columns.
5. Encode categorical features using one-hot encoding, ordinal encoding, target encoding, or another appropriate method.
6. Extract useful information from date fields such as `host_since`, `first_review`, and `last_review`.
7. Engineer text-based features from listing descriptions, amenities, and house rules.
8. Train a regression model.
9. Generate predictions for `test.csv`.
10. Save predictions in the required submission format.

Potential models include linear regression, ridge regression, random forests, gradient-boosted trees, XGBoost, LightGBM, CatBoost, or neural models. Stronger submissions will likely combine careful cleaning, location features, categorical encoding, and robust handling of missing values.

Evaluation

Submissions are evaluated by comparing predicted prices against the hidden true prices for the test set.

Your goal is to minimize prediction error. Because this is a price prediction problem, models should avoid producing negative prices. If your model outputs negative values, clip them to zero or use a target transformation that ensures nonnegative predictions.

Citation

Ben TenWolde. UCSD Spring 2026 DSC148: Data Mining. https://kaggle.com/competitions/ucsd-spring-2026-dsc-148, 2026. Kaggle.

Cite

Competition Host
BenTenWolde

Prizes & Awards
Kudos
Does not award Points or Medals

Participation
31 Entrants
8 Participants
8 Teams
22 Submissions

Tags
Mean Absolute Error

Table of Contents

Overview
Description
Evaluation
Citation